



蚂蚁开源
ANT OPEN SOURCE



蚂蚁开源技术沙龙

面向万级 SWE 任务 Sandbox、 镜像分发与 RL 工程实践

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SWE Is the Foundation

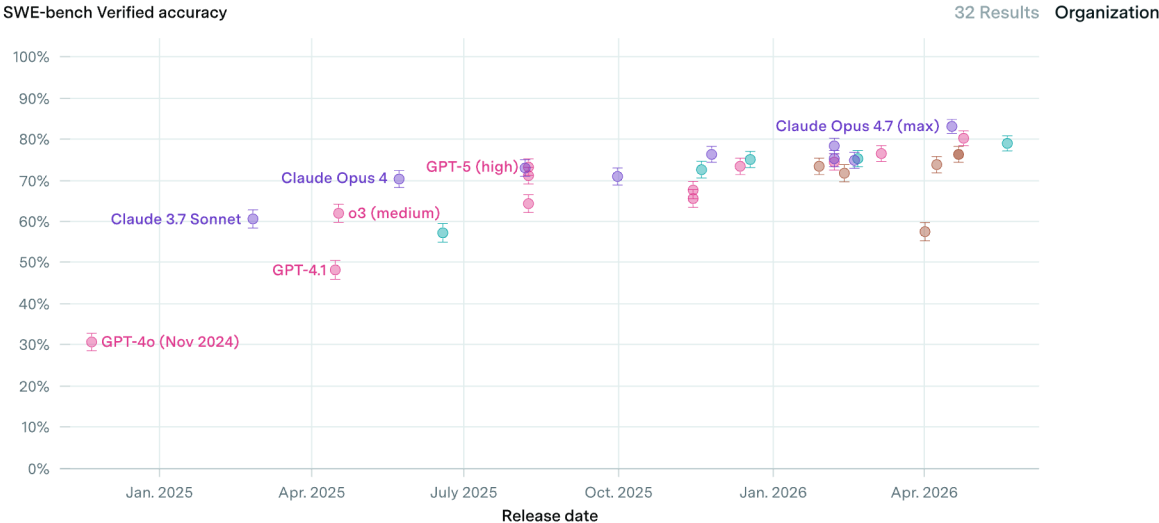


Claude Code



Codex

AI performance solving issues from 12 open-source Python repositories



Evaluation

real repo repair is the signal

SWE-Bench

real python issues

SWE-Bench Pro

professional-level

SWE-Bench Multilingual

Multi-language

What Is a SWE Task?

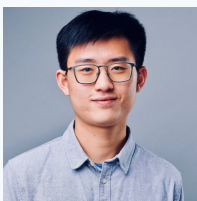
astropy / astropy #12907

real issue

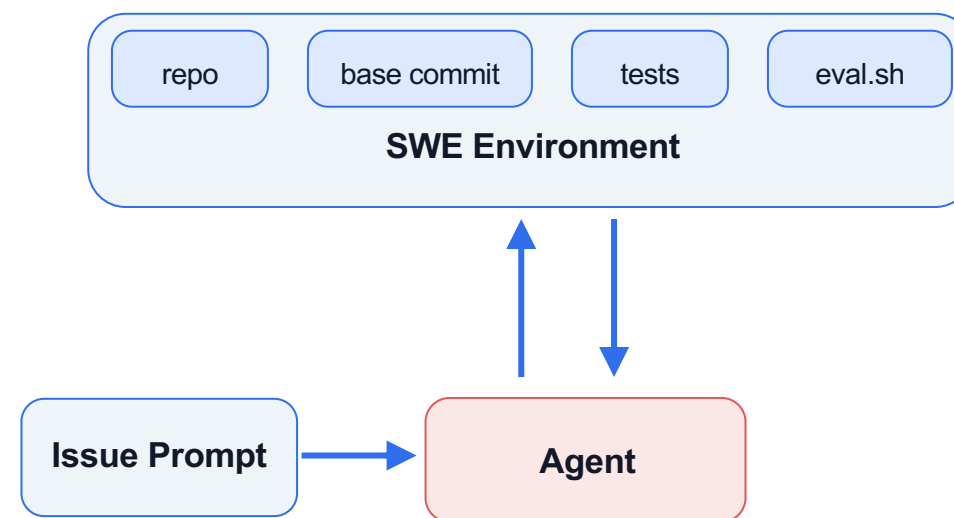
Model separability matrix

nested model result is wrong

```
from astropy.modeling import models
separability_matrix(compound_model)
expected diagonal, got nested mismatch
```

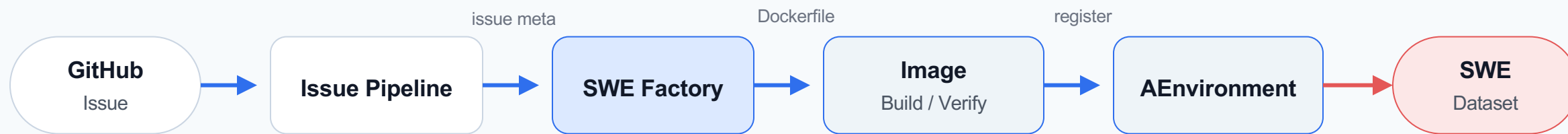


Full development context



The environment is part of the task

From GitHub Issues to SWE Environments



RL inputs

Real swe environment
Language diversity
Repo diversity

Builds
20K+ images

layered Dockerfiles
fail2pass verified

Coverage
3K+ repos

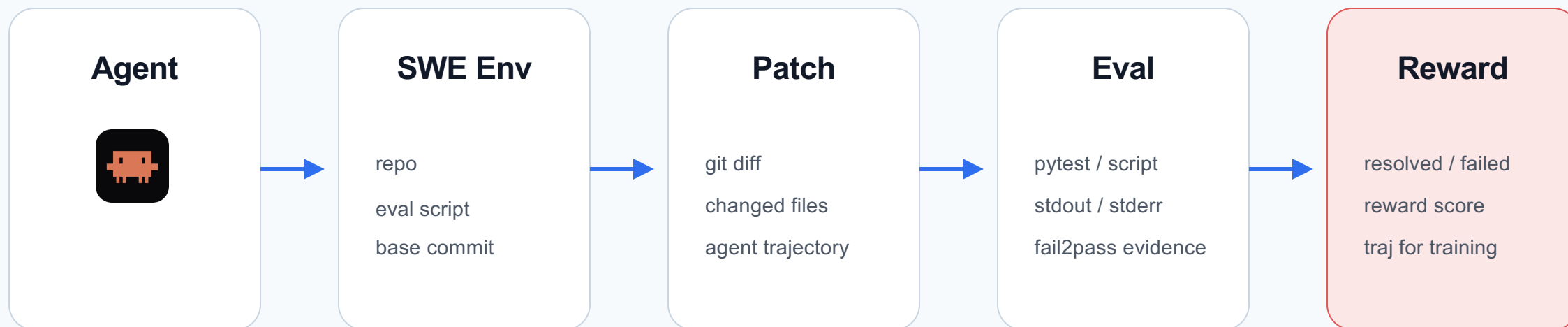
Python / Java / Go
more langs

Environment assets

image / deploy config
issue prompt / eval.sh

Scaling SWE RL means scaling reproducible software environments

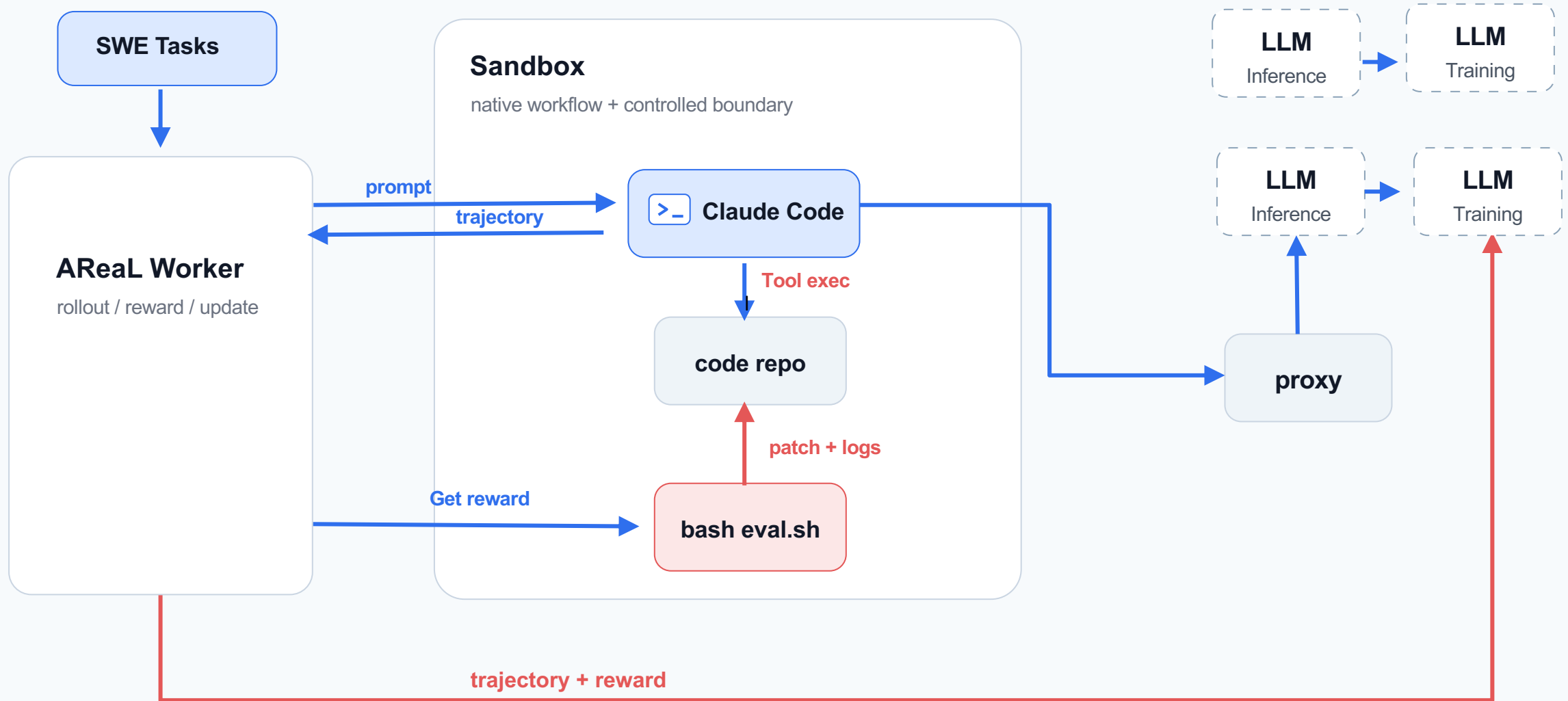
From SWE Environment to Reward



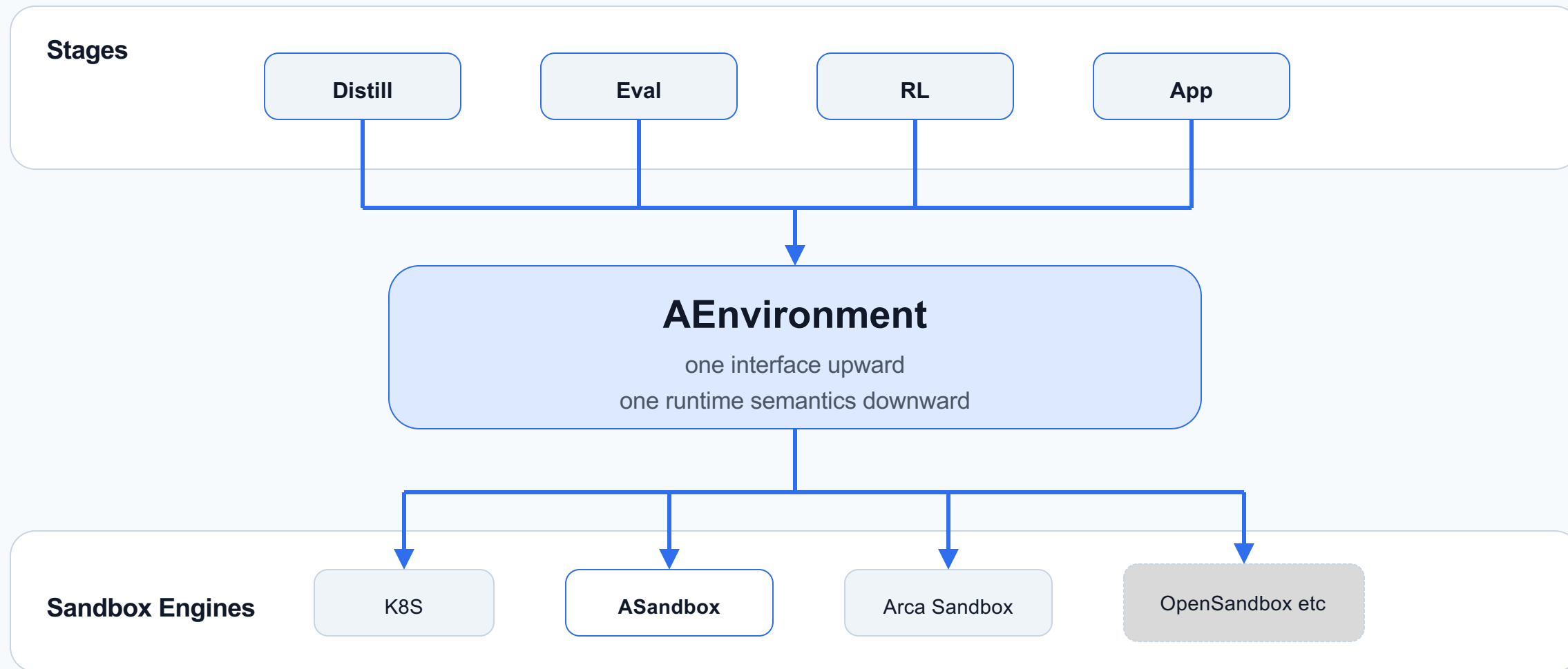
Reward is not a static label

It is the execution result inside the same environment.

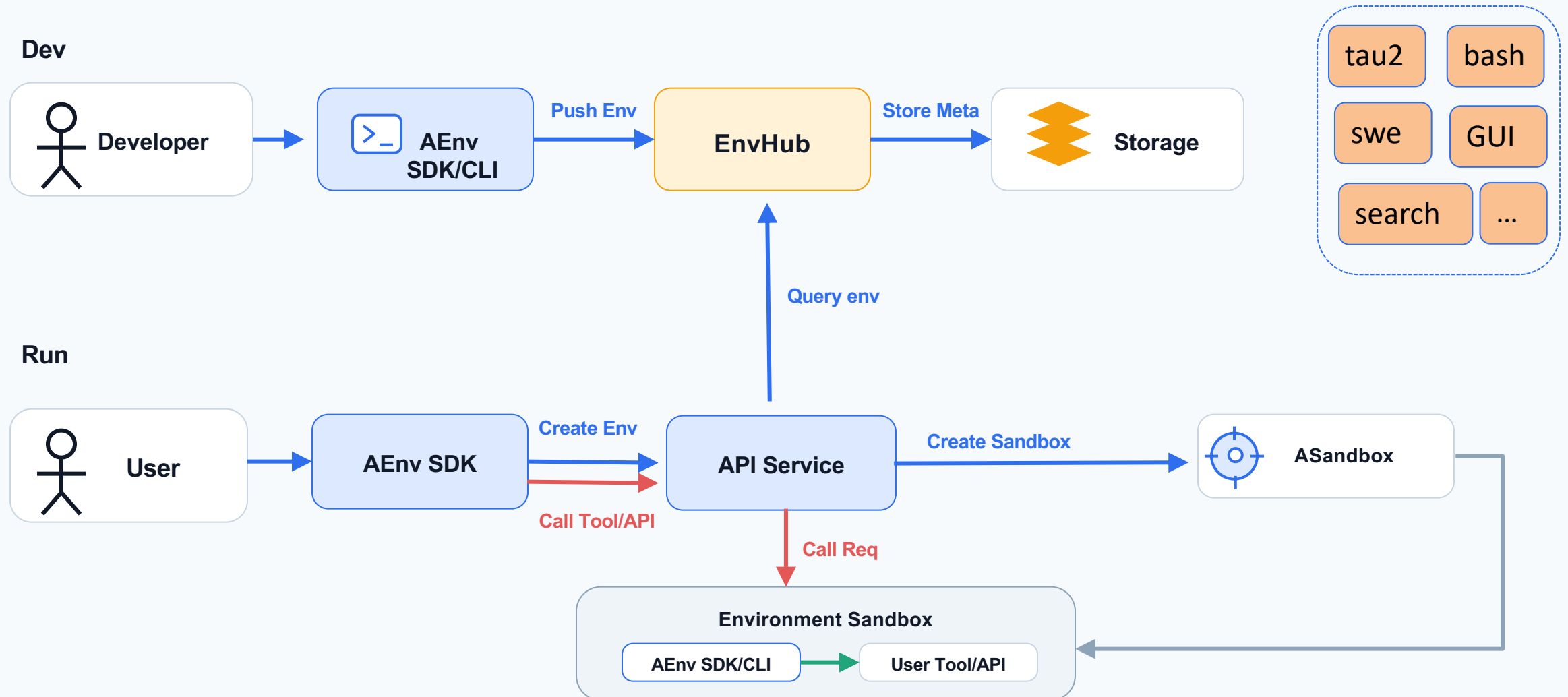
RL with SWE Environment



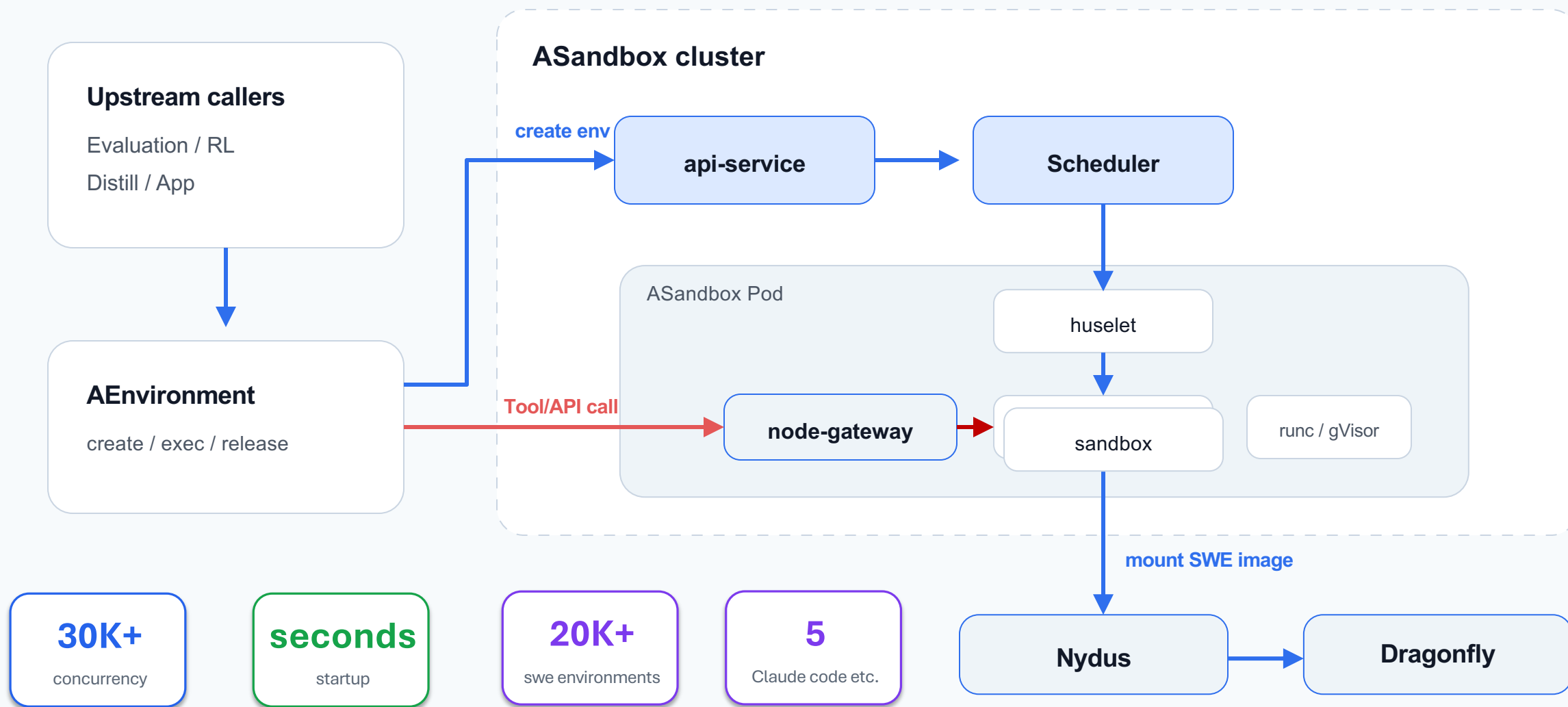
Consistent Environments Across Stages



AEnvironment: Unified Environment Interface

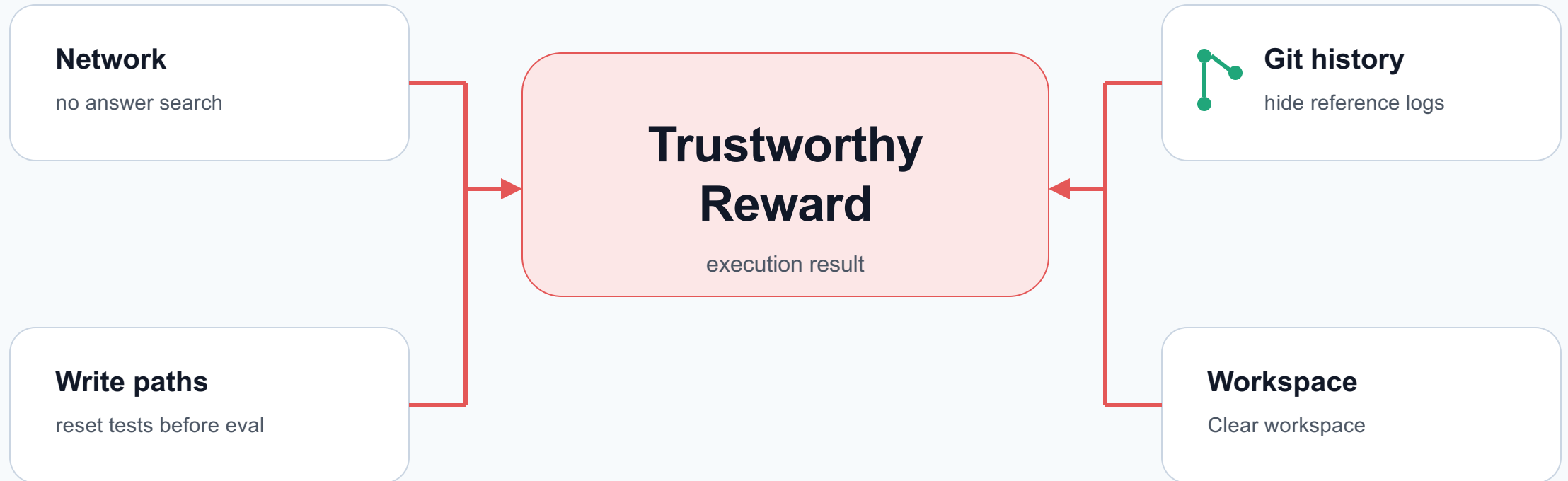


ASandbox: Stateful Sandbox Engine



When Boundaries Are Unclear

Unclear boundary: agent optimizes loopholes

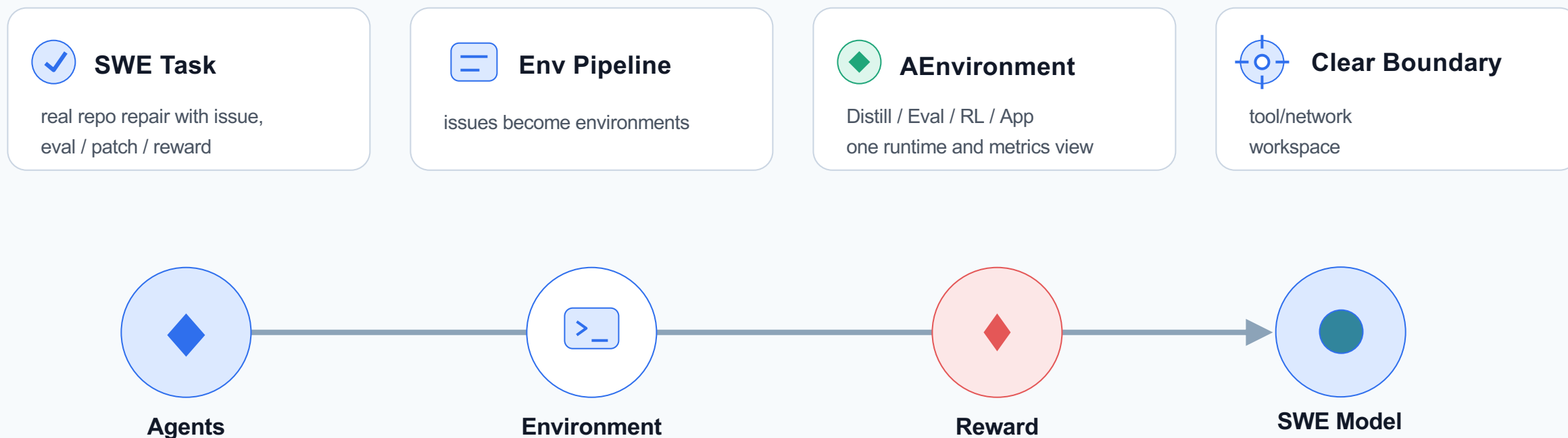


Clear boundary: real reward

Summary

Scaling SWE RL Means Scaling Environments

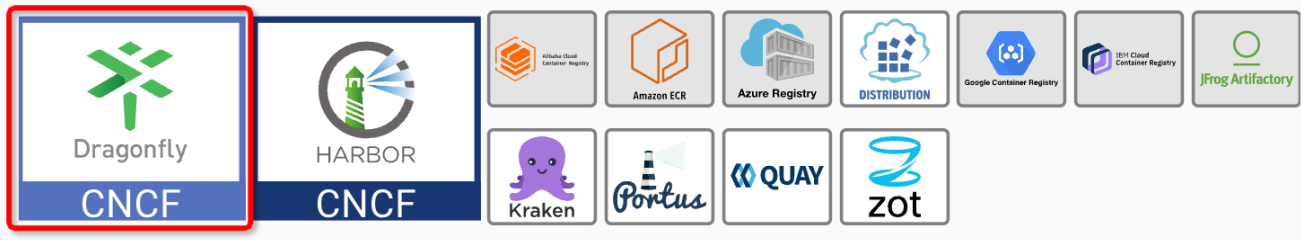
Strong agents need reproducible, observable, recoverable environments.



Dragonfly



Container Registry



Container Runtime



What is Dragonfly?

Provide efficient, stable and secure file distribution and image acceleration based on p2p technology to be the best practice and standard solution in cloud native architectures.

It is hosted by the Cloud Native Computing Foundation(CNCF)as an *Incubating* Level Project.

There are more than *100* contributors, and maintainers come from *Ant Group, Alibaba Group, ByteDance, Intel, Baidu Group, Zhipu AI* and *Dalian University of Technology*.

Public cloud users include *Alibaba Cloud(Aliyun), Google Cloud Platform (GCP), Volcano Engine, Baidu AI Cloud*, etc.



Dragonfly

Image Acceleration

Dragonfly supports various container clients such as *containerd*, *Docker*, *cri-o*, *ORAS*, etc. It provides three solutions for image acceleration.

The first solution is to use Dragonfly to distribute images based on P2P technology, which is suitable for *large-scale cluster*.

The second solution is to use Dragonfly and Nydus to distribute accelerated images, which is suitable for *large-scale cluster* and *faster container launching*.

The third solution is to use Nydus to distribute accelerated images, which is suitable for *faster container launching*.

File Distribution

Dragonfly supports large-scale file distribution and uses P2P technology to *eliminate the impact of origin bandwidth limitations*.

It supports file distribution of protocols including *HTTP*, *HDFS*, etc. Additionally, it also supports different object storage protocols includes *S3*, *OSS*, *OBS*, etc.

Add *Dfstore* to expand the file distribution capability, it can depend on different types of object storage, such as S3, OSS, OBS, etc. to *provide stable object storage capabilities*.

AI Infrastructure

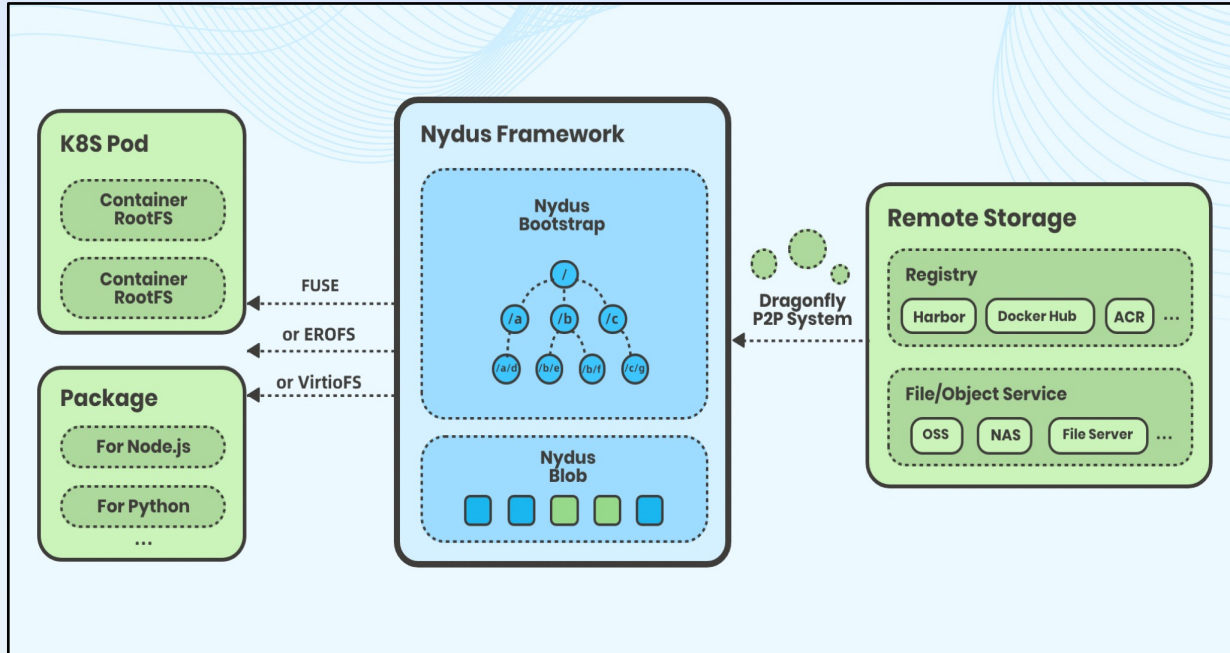
Dragonfly supports distributing data during *AI training* and *AI inference*.

In the AI inference, Dragonfly supports *Triton Server* and *TorchServe* to use Dragonfly distribution model.

In the AI training, *Fluid* downloads dataset through Dragonfly when running based on *JuiceFS*.

Supports downloading models and datasets from *Hugging Face Hub* by SDK through Dragonfly HTTP proxy.

Nydus



What is Nydus?

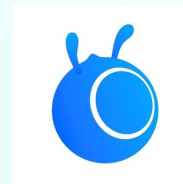
Provide a content-addressable file system on the RAFS format, which enhances the current OCI image specification by improving container launch speed, image space and network bandwidth efficiency, and data integrity.

It is a *sub-project of Dragonfly*. It can reduce the end-to-end cold start time of containers from *minutes to seconds*, and supports the creation of *millions of containers* every day in production.

Maintainers come from *Ant Group*, *Alibaba Group*, *ByteDance*, *DataDog*, etc.

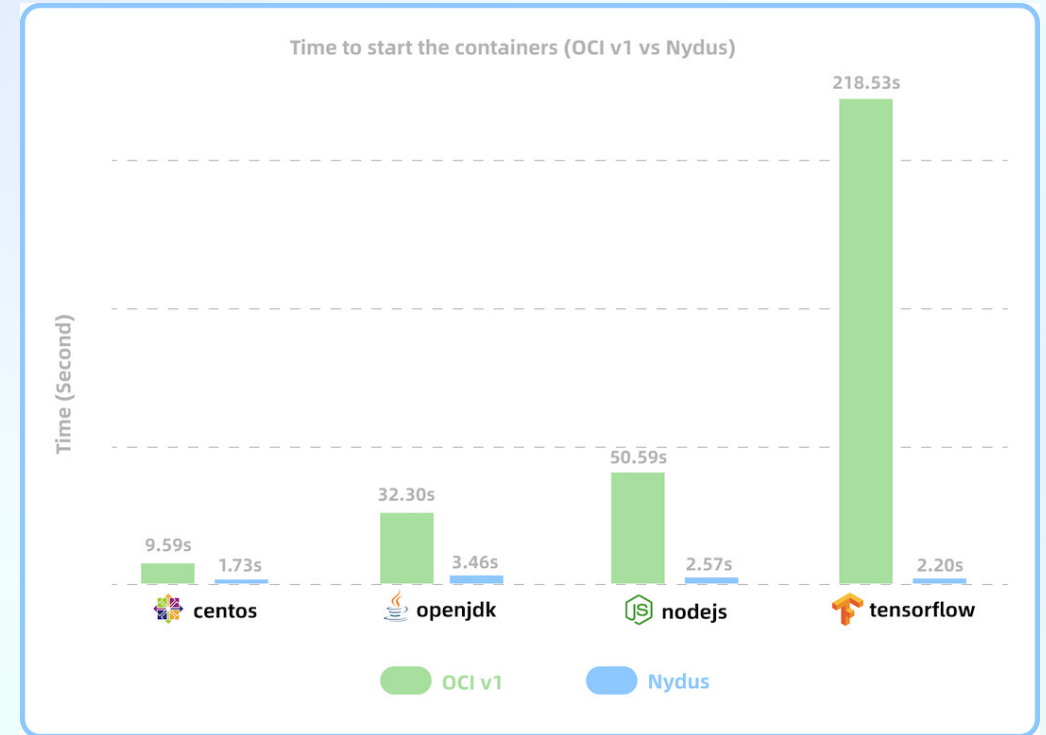
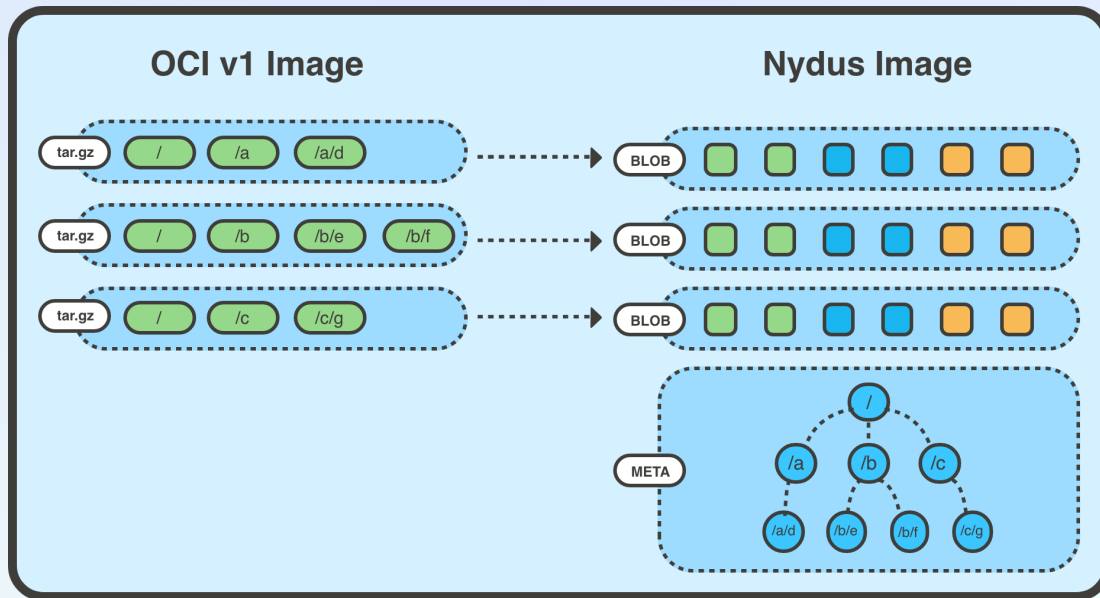


CLOUD NATIVE
COMPUTING FOUNDATION



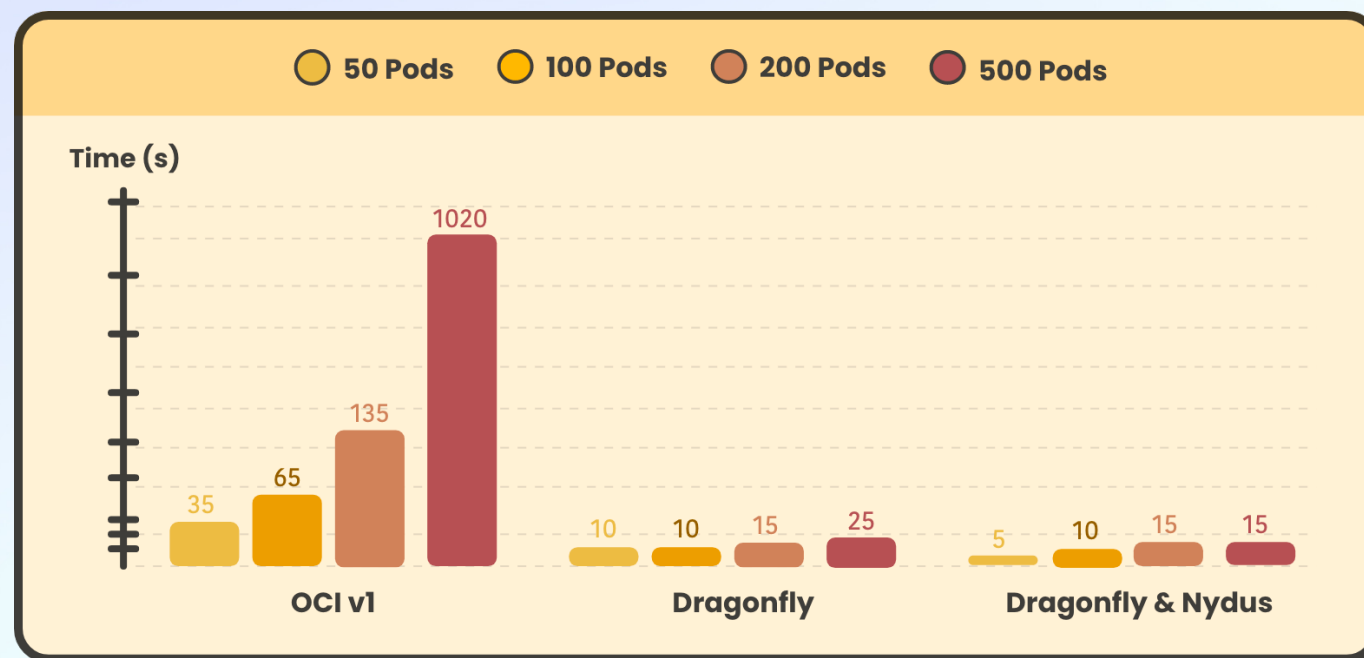
Nydus

On-demand Load & Chunk Deduplication



Benchmark

镜像分发：分钟级到秒级



Nginx Pod Creation to Container Start

SWE场景

镜像规模大，任务要求长稳

镜像规模：镜像数量 10w+ 量级，6k 分组，最大分组为 1k。

预热挑战：10w 镜像约 150TiB，单 seed client 节点无法全量预热。

去重收益：同组去重率 70%+，启动阶段 ondemand 数据 99% 在公共 chunk 内。

按需收益：5GiB 镜像，启动阶段 ondemand 数据读取量约 100MiB。

Agent Sandbox 场景

秒级启动，任务快照等需求

分发挑战：按需加载 FUSE 不开启预取 3m4s，开启 prefetch 1m12s。

按需收益：镜像总大小 2.5GiB，E2E 启动只需要 50MiB。

内存与 RootFS 快照：内存快照 3.6GiB 数据，压缩后为 524MiB。